

Stack Overflow Developer Survey Analysis 2025

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OUTLINE



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- Methodology
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 - Visualization – Charts
 - Dashboard(Current Usage , Future Trends Demographics.
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- Conclusion
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Job posting Analysis

EXECUTIVE SUMMARY



- **Point 1:**-Python, JavaScript, and SQL remain the most used programming languages.
- **Point2:**-Implications: Technology adoption is shifting toward
 - Sub Point 1:- **modern**
 - Sub Point 2:- **scalable**
 - Sub Point 3:-**cloud-oriented solutions.**
- **Point3:**-Rust and Go show strong future growth trends.
- **Point4:**-SQL, PostgreSQL, and MySQL dominate current databases.
- **Point5:**-Cloud-native databases (MongoDB, Firebase) expected to grow.

INTRODUCTION



- Point1:-**Purpose:** To analyze current and future technology trends using developer survey data.
- Point2:-**Target Audience:** Technology managers, HR professionals, and developers.
- Point3:-**Value:** Helps organizations align hiring, training, and investment strategies with market trends.
- Point4:- **Value:**
 - Sub Point1:- Helps organizations align hiring, training,
 - Sub Point2:-investment strategies with market trends.



METHODOLOGY



- *Data source: Stack Overflow Developer Survey 2025*
- *Data wrangling: Handled missing values Split multi-answer columns (;)*
- *Aggregated top 10 programming languages & databases*
- *Visualizations created using Python (pandas, matplotlib, openpyxl)*
- *Dashboard created with Looker Studio / Cognos*

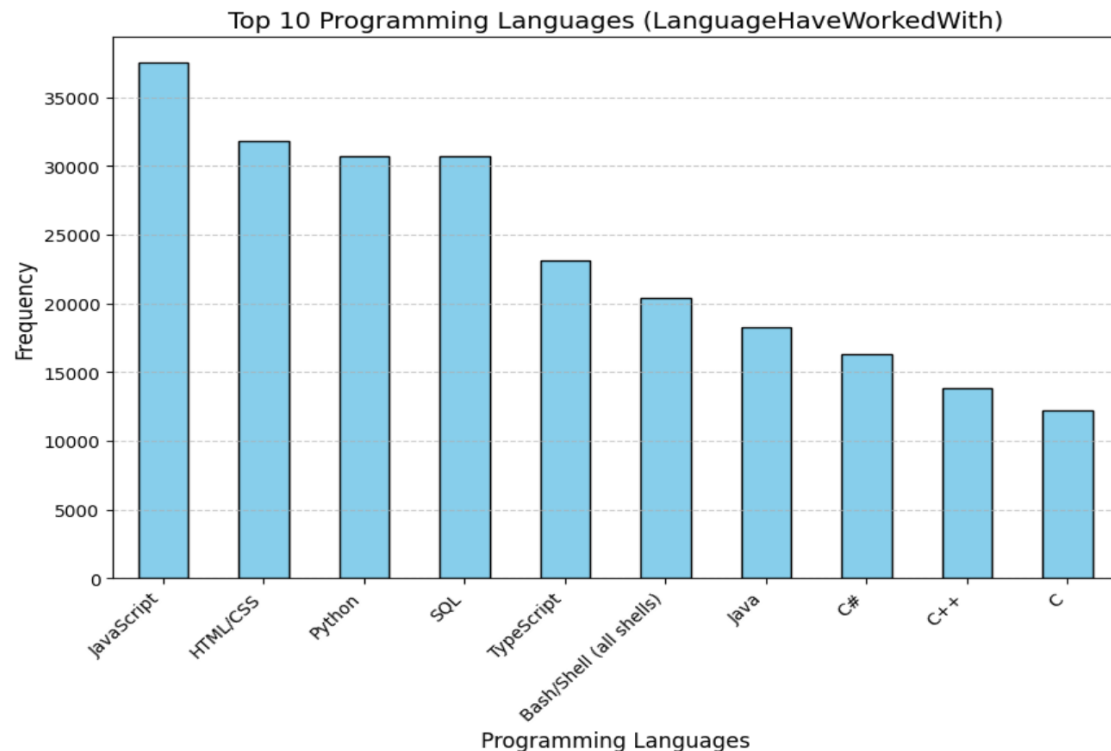
RESULTS

- PROGRAMMING LANGUAGE TRENDS
- PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS
- DATABASE TRENDS
- DATABASE TRENDS - FINDINGS & IMPLICATIONS

PROGRAMMING LANGUAGE TRENDS

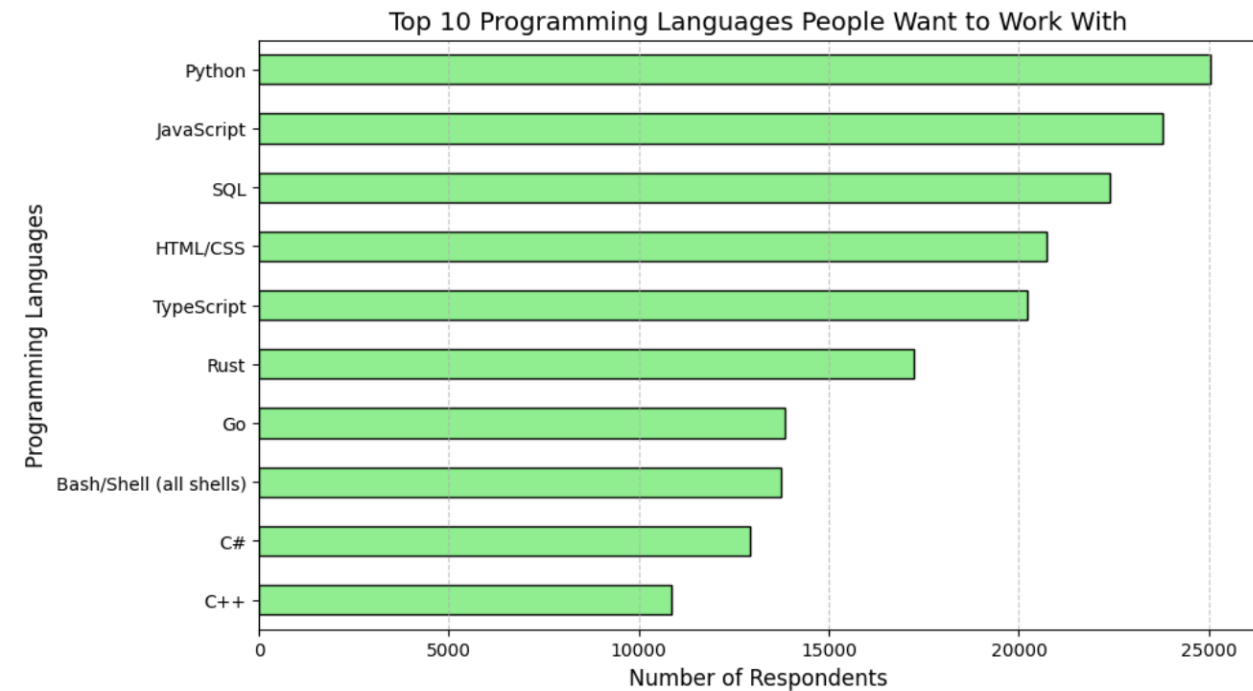
Current Year

<Bar chart of top 10 programming languages for the current year goes here.>



Next Year

< Bar chart of top 10 programming languages for the next year goes here.>



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

- **Current (Have Worked With)**

- **Top 3:** JavaScript, Python, SQL
- JavaScript still widely used for web dev
- Python dominates in **AI/ML, data science**
- SQL remains essential for backend and analytics

- **Future (Want to Work With / Projection)**

- **Python** continues strong — developers want to deepen skills
- **TypeScript** shows significant growth (safer alternative to JS)
- **Go** and **Rust** gaining momentum for system programming & performance-critical apps

- **Implication:** Companies can expect hiring demand in Python, TypeScript, and Go skills.

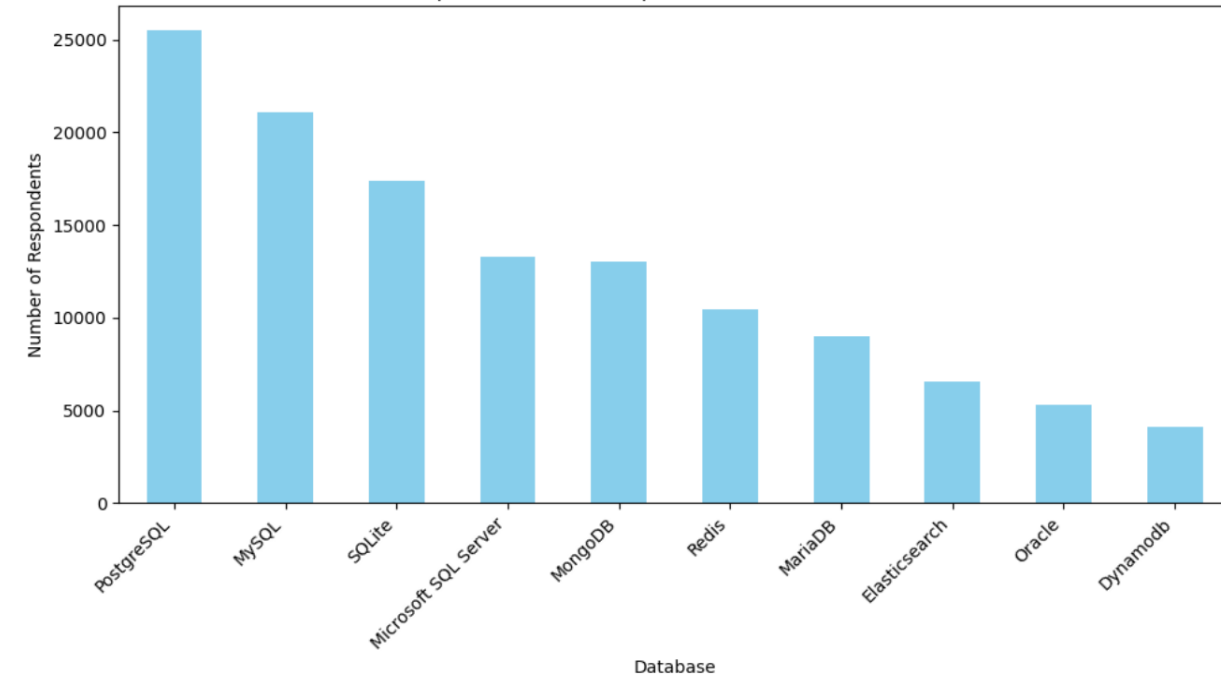


DATABASE TRENDS

Current Year

< Bar chart of top 10 databases for the current year goes here >

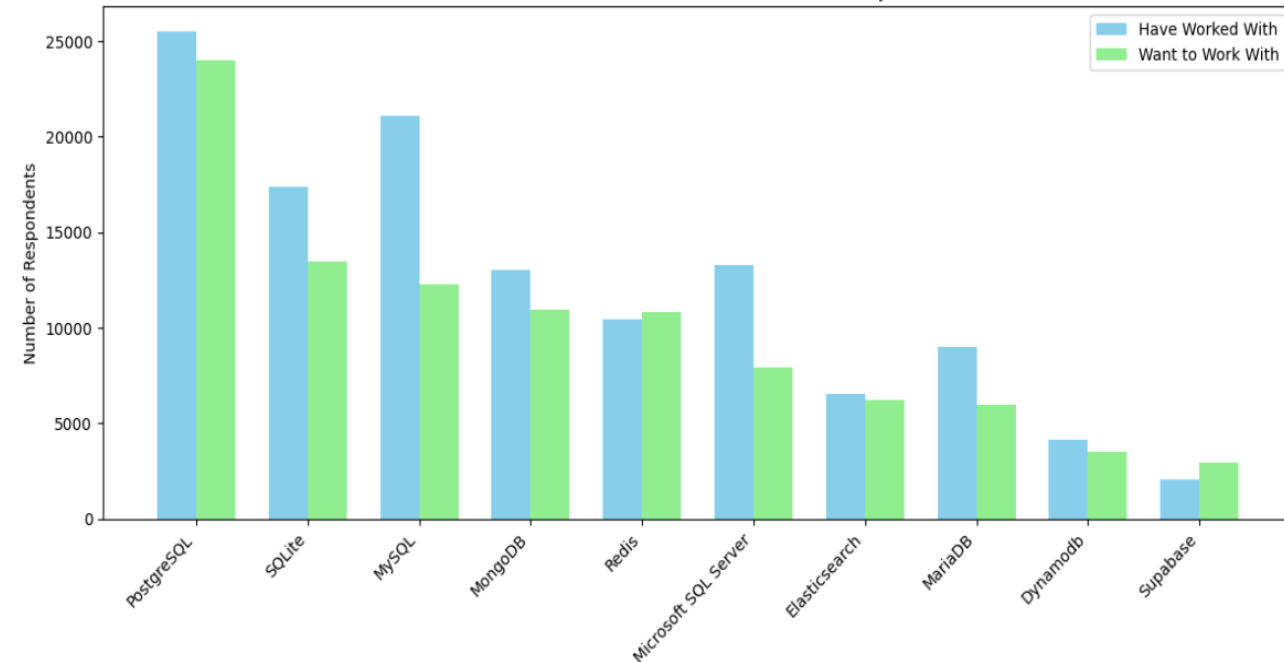
Top 10 Databases Respondents Have Worked With



Next Year

< Bar chart of top 10 databases for the next year goes here.>

Database Skills: Current vs Future Demand Projection



DATABASE TRENDS - FINDINGS & IMPLICATIONS

- **Current (Have Worked With)**

- **Top 3:** MySQL, PostgreSQL, Microsoft SQL Server
- MySQL remains widely used due to legacy systems
- PostgreSQL gaining adoption thanks to open-source flexibility

- **Future (Want to Work With / Projection)**

- **PostgreSQL:** highest projected demand
- **MongoDB:** popular for NoSQL, scalable apps
- **Redis:** rising for caching and high-performance apps

- **Implication:** Future demand shifts toward **modern, flexible, cloud-friendly databases** (PostgreSQL, MongoDB).



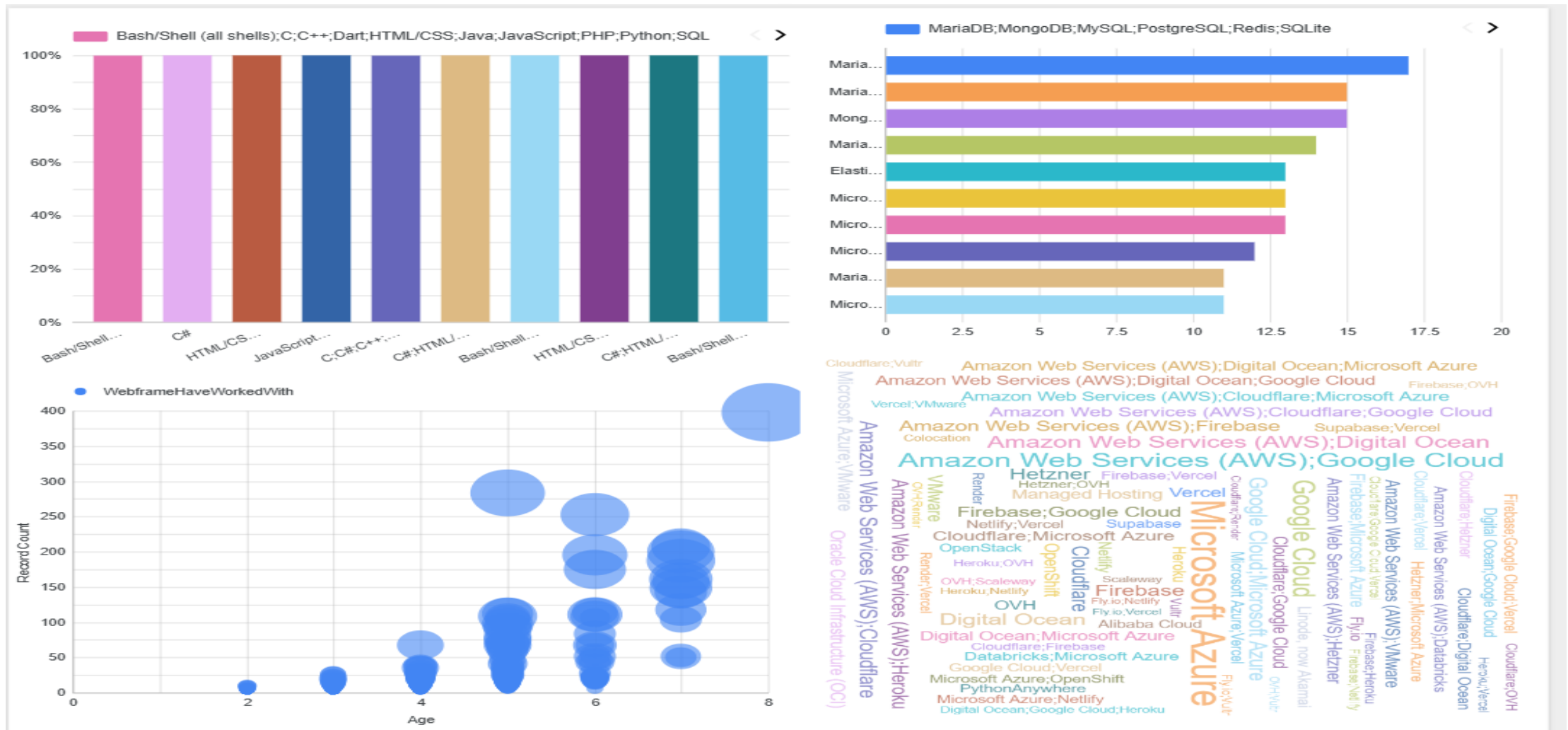
DASHBOARD



<https://lookerstudio.google.com/reporting/26ea8cc3-80db-4b9b-9501-108ec5512b09>



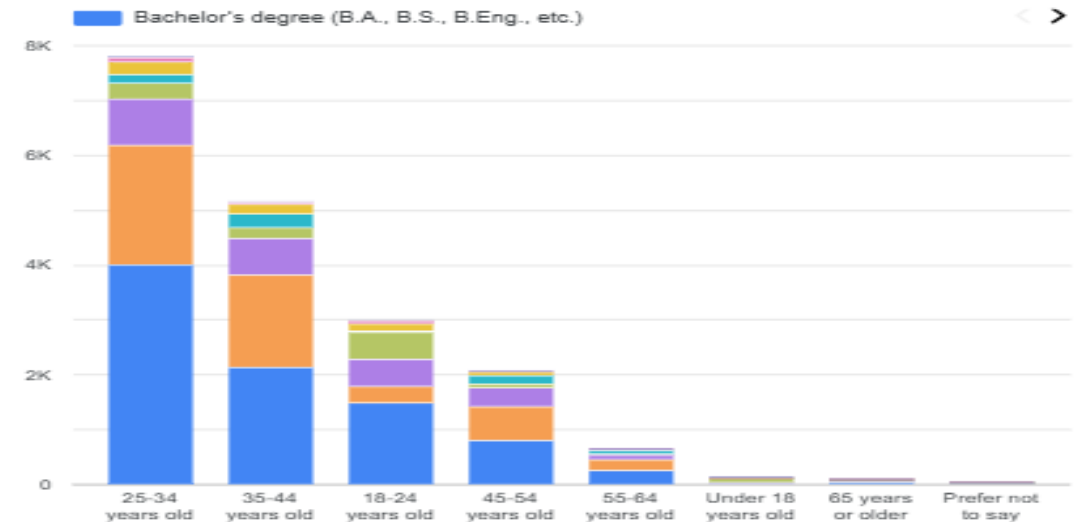
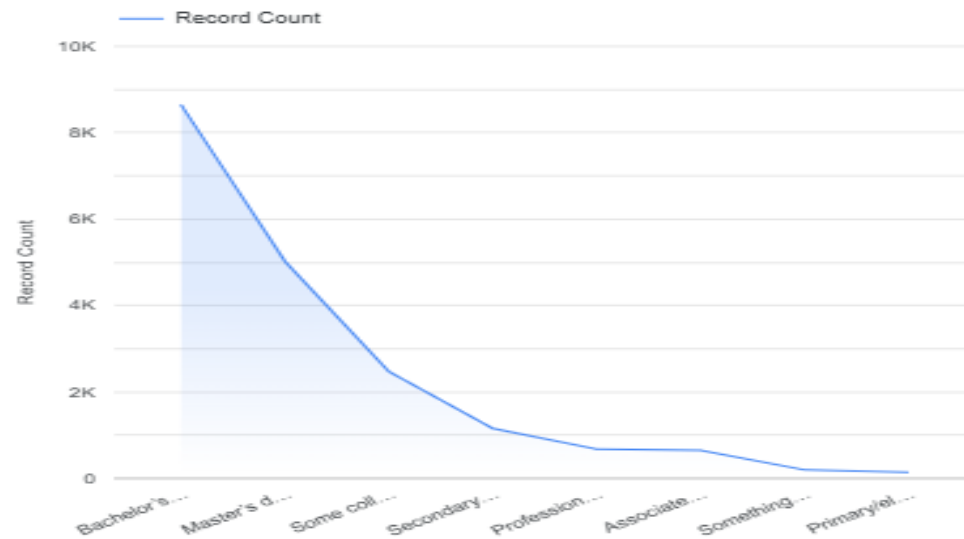
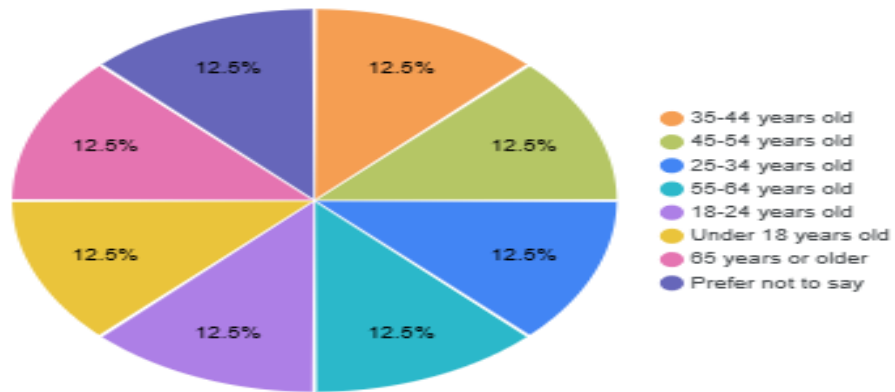
Current Technology Usage



Future Technology Trends



Demographics



Dashboard Insights

- Bullet points summarizing trends from dashboards:
- Python and TypeScript rising
- PostgreSQL and MongoDB high demand
- Regional tech adoption varies

DISCUSSION



- **Bullet points summarizing trends from dashboards:**
- Python and TypeScript rising
- PostgreSQL and MongoDB high demand
- Regional tech adoption varies

OVERALL FINDINGS & IMPLICATIONS

- **Python & TypeScript** → clear winners for future programming languages.
- **PostgreSQL & MongoDB** → leading databases for future demand.
- Job postings confirm strong alignment with survey projections (Python + PostgreSQL).
- Industry trend: Shift toward **cloud, open-source, and performance-optimized stacks**.

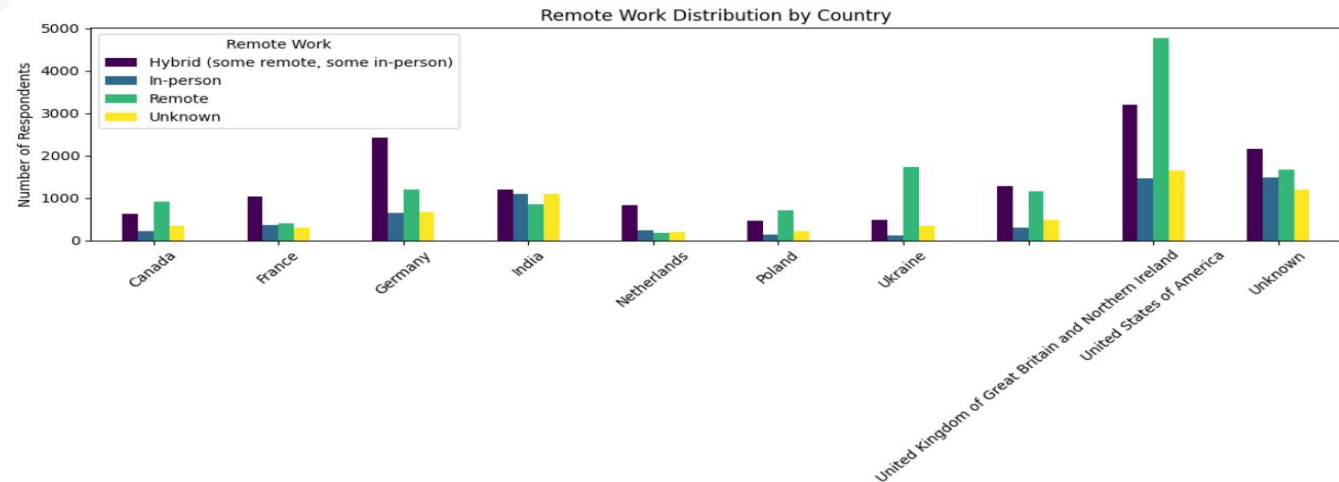
CONCLUSION



- Point 1:-Python, SQL, and JavaScript remain dominant.
- Point 2:-Rust, Go, and cloud databases are **future-ready technologies**.
- Point 3:-Demographics suggest growing global participation in development.
- Point 4:-Organizations should invest in **continuous training & hybrid skills**.

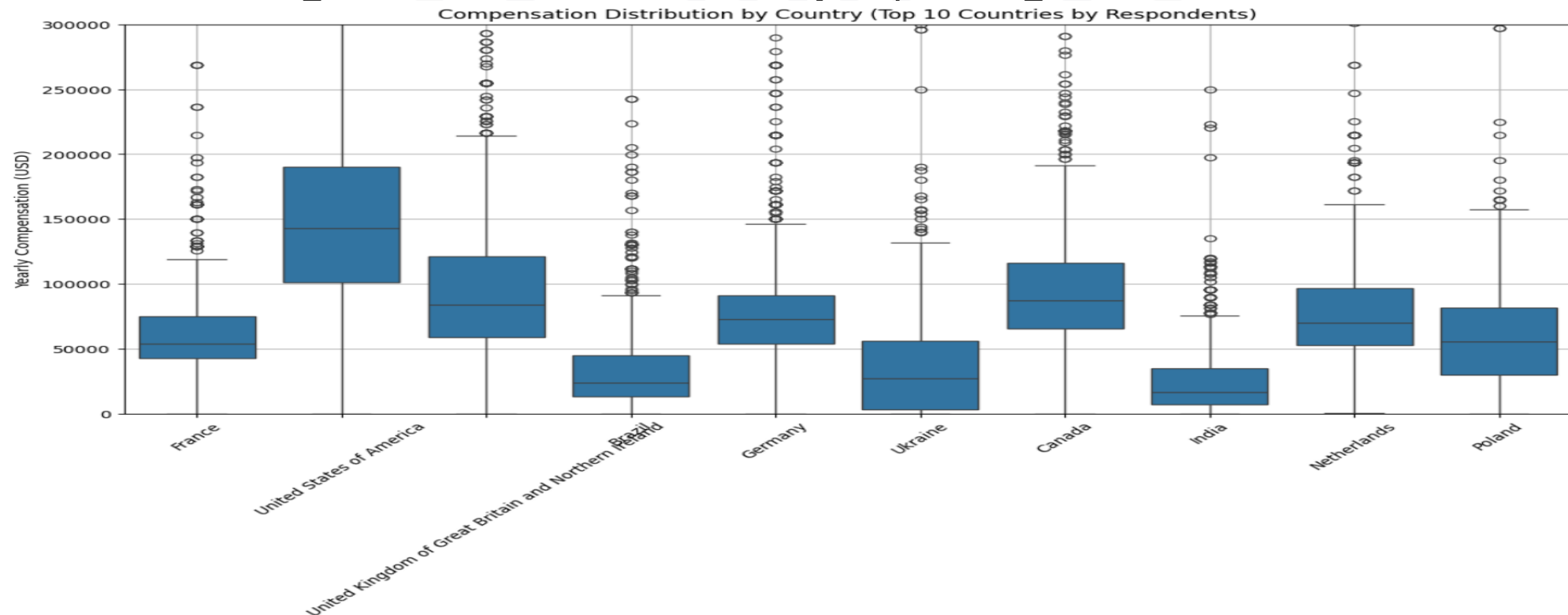
APPENDIX

- Include any relevant additional charts, or tables that you may have created during the analysis phase. Yes, created.



JOB POSTINGS

In Module 1 you have collected the job posting data using Job API in a file named “job-postings.xlsx”. Present that data using a bar chart here. Order the bar chart in the descending order of the number of job postings.



POPULAR LANGUAGES

In Module 1 you have collected the job postings data using web scraping in a file named “popular-languages.csv”. Present that data using a bar chart here. Order the bar chart in the descending order of salary.

